

Foundations • Case Studies • Portfolio Theory • Reading List

A learning document, generalized from a personal study guide. Not investment advice.

Long-Term U.S. Markets

A 30-Year Curriculum (1996–2026)

1. Introduction & How to Use This Guide

This curriculum is written for a young investor with a 40+ year time horizon, investing through a Roth IRA and a taxable brokerage account, with a portfolio tilted toward technology and growth equities — an S&P 500 index-fund core, a concentrated set of individual technology stocks (semiconductors, a mega-cap platform, a cybersecurity name), and a small allocation to a spot Bitcoin fund — and an interest in expanding into full portfolio theory and factor investing. The goal of this document is not to tell you what to buy. It is to help you read markets across multiple timeframes, understand why prices have moved the way they have over the past 30 years, and build the analytical and behavioral framework to keep compounding for the next 40+.

How to use this guide

Read it in order the first time. Each chapter assumes the foundations from earlier chapters.

Pair every chapter with the interactive 30-year chart artifact. The annotated events on that chart correspond directly to the case studies in Chapter 6.

Don't skip the behavioral chapters (11 and 12). Risk-adjusted returns are won and lost there, not in stock picking.

Treat the reading list (Chapter 14) as an open invitation, not a checklist. Pick the books that map to where you currently feel weakest.

Source-of-truth note

The numerical claims, performance figures, and historical episodes in this document are drawn from a combination of widely cited academic and industry sources, including the S&P Dow Jones Indices SPIVA scorecards, Morningstar fund research, Robert Shiller's historical dataset, Jeremy Siegel's Stocks for the Long Run datasets, the Federal Reserve's FRED database, and Vanguard's and Fidelity's public research. Where I quote a specific statistic (e.g., 85–90% of active funds underperform over 15 years), the underlying source is named so you can verify it yourself. This is a learning document, not investment advice.

2. A Starting-Position Framework

This section is the analytical lens the rest of the curriculum is written through: how to read a portfolio's strengths and risks before layering on 30 years of history. It is framed around a common profile — a young investor with a multi-decade horizon, contributing through a Roth IRA and a taxable brokerage, holding roughly 95% equities (an S&P 500 index-fund core plus a concentrated set of individual technology stocks) and a small (~5%) spot-Bitcoin position.

What is structurally strong about a profile like this

A multi-decade horizon. The single most powerful variable in investing is time. A dollar compounding at 8% real per year becomes about 21x in real terms over 40 years and about 69x over roughly 55 years. Even partial use of that runway dwarfs almost any other lever.

Roth IRA capacity early. A Roth grows and is withdrawn tax-free. Over 40+ years of compounding it is mathematically the most valuable account most people will ever own; maxing it in every eligible year is the sensible default.

High human capital, low fixed expenses. Decades of expected earnings are the largest asset on the balance sheet. Low fixed costs mean an investor rarely has to sell during a drawdown — the most common way people permanently destroy wealth.

What is genuinely risky about it

Concentration in semiconductors. NVDA, AMD, and AVGO are all semiconductor companies. Semis are historically the most cyclical large sector — the Philadelphia Semiconductor Index has drawn down 80%+ (2000–2002), 60%+ (2008–2009), and 35%+ (2022). That is a category fact, not a forecast. Chapter 7 is dedicated to it.

Untested risk tolerance. An investor whose real-money experience began after 2022 has not lived through a sustained drawdown. The first one is psychologically the hardest. Chapter 11 is built for this.

A high individual-stock weight. A ~75% allocation to single names can pay off enormously and can also magnify drawdowns. Diversification is not a guarantee of higher returns; it is a guarantee of lower variance for a given expected return. The right number depends on an honest answer to: at what drawdown do you change your behavior?

A small crypto position. Bitcoin has had four >70% drawdowns since 2013. A ~5% allocation, sized so a total wipeout would not change the long-run plan, is defensible; a 5% allocation that would cause panic at \$0 is not. Chapter 8 covers this.

How this curriculum is calibrated

Examples preferentially reference the kinds of names in this profile — semiconductors (NVDA, AMD) in the cycle chapter, mega-cap platforms (AMZN) in the platform chapter, cybersecurity (CRWD) in

the software chapter, an S&P 500 index fund (SWPPX) in the index chapter, and a spot Bitcoin ETF (FBTC) in the crypto chapter — because concrete names make the history stick.

3. The Four U.S. Indices — What They Actually Measure

Most investors use “the market” as a single word and never separate the four most cited U.S. indices. They measure different things, are weighted differently, and tell you different stories about the economy. You need to be able to read all four.

3.1 The Dow Jones Industrial Average (Dow / DJIA)

Created by Charles Dow in 1896, the Dow tracks 30 large U.S. companies hand-picked by an editorial committee at S&P Dow Jones Indices. It is the oldest continuously published equity index in the world and the one cited most often in mainstream news (“the Dow is up 200 points”), but it is also the least useful as a market benchmark for serious investors.

Why the Dow is structurally weird

Price-weighted. The Dow weights companies by their share price, not by company size. A \$400-per-share stock has more index weight than a \$40-per-share stock even if the \$40 stock is a much larger company. This is an accident of 1896 arithmetic that has never been corrected.

Only 30 companies. With 30 names, a single company’s move can swing the index more than the broad economy.

Hand-curated. Companies are added and removed by committee. Historically the committee has been slow to add tech (Apple was not added until 2015) and quick to remove laggards, which builds in survivorship bias.

How to use the Dow

Use the Dow as a cultural and historical reference, not as an analytic benchmark. When the headline says “the Dow,” translate it in your head to “a price-weighted basket of 30 large U.S. companies that does not necessarily reflect the broader market.” For everything else, use the S&P 500 or a total market index.

3.2 The S&P 500

The S&P 500 tracks the 500 largest U.S. companies that meet a set of inclusion criteria (profitability, market cap, liquidity, U.S. listing). It is market-cap-weighted: bigger companies count more. It captures roughly 80% of total U.S. equity market capitalization. This is the index serious investors and academics use as the U.S. equity benchmark.

Why the S&P 500 is the default benchmark

Broad coverage with high quality. 500 companies, 11 sectors, weighted by economic significance.

Methodologically clean. Inclusion is rules-based (with some committee discretion at the margin), and methodology is published.

Investable. Trillions of dollars track the S&P 500. SWPPX is the Schwab S&P 500 Index Fund, expense ratio 0.02%. Equivalent products exist at Vanguard (VFIAX, VOO) and Fidelity (FXAIX).

What the S&P 500 hides

The cap-weighting that makes the S&P 500 a clean benchmark also concentrates it. As of 2025, the top 10 companies (mostly tech mega-caps) are roughly 35–38% of the index. When you buy SWPPX, you are buying a market-weighted bet on what is currently the largest set of companies. That is a feature for long-term compounding (winners stay winners until they don't) but it is also a concentration risk that did not exist to the same degree 25 years ago.

3.3 The NASDAQ Composite and NASDAQ-100

“The NASDAQ” in casual conversation usually refers to one of two distinct indices: the NASDAQ Composite (~3,000+ stocks listed on the NASDAQ exchange) and the NASDAQ-100 (the 100 largest non-financial stocks on the NASDAQ exchange, the index that the QQQ ETF tracks). Both are heavily tilted toward technology, but they are not identical.

Composition tilt

Roughly 50–60% technology, depending on how you classify mega-caps like Amazon (consumer discretionary by GICS, but functionally tech).

Heavy in semiconductors, software, internet platforms, and biotech.

Almost no traditional financials, energy majors, or industrials.

Why this matters here

A concentrated individual-stock book of this kind is essentially a concentrated NASDAQ-100. NVDA, AMD, AVGO, CRWD, and AMZN are all NASDAQ-listed. SWPPX (S&P 500) gives you broader exposure. Understanding the NASDAQ's history — specifically the 1995–2002 boom-bust and the 2022 drawdown — is non-negotiable for an investor with this concentration.

3.4 The NYSE Composite and “The NYSE”

The New York Stock Exchange is an exchange (a venue), not an index. The NYSE Composite is a less-cited cap-weighted index covering all common stocks listed on the NYSE. It is heavier in financials, energy, industrials, and consumer staples than the NASDAQ Composite, which makes it a useful complement when you want to know what “old-economy” America is doing.

NYSE vs NASDAQ as venues

NYSE: traditionally a designated-market-maker (specialist) auction model on a physical floor; today mostly electronic with floor presence. Historically home to large industrials, banks, energy, and blue

chips.

NASDAQ: founded 1971 as the world's first electronic stock market. Dealer-based, fully electronic. Became the home of high-growth tech.

In practice, where a company lists is more about brand and history than about how its stock trades day-to-day. Berkshire Hathaway and JPMorgan are NYSE-listed; Microsoft, Apple, Nvidia, and Amazon are NASDAQ-listed.

3.5 Other indices worth knowing

Russell 2000: the 2,000 smallest stocks in the Russell 3000. The standard small-cap benchmark. Useful for tracking when the market broadens beyond mega-caps.

Russell 1000: the 1,000 largest. A broader large-cap proxy.

Wilshire 5000 / CRSP US Total Market: ~total market indices. VTSAX and FZROX track total market funds.

MSCI EAFE / MSCI ACWI ex-US: developed and total international ex-US equity benchmarks.

4. Reading Long-Term Charts

Most retail investors lose money not because they pick bad stocks but because they read charts wrong. This chapter is the toolkit.

4.1 Linear scale vs logarithmic scale

The single most important chart-reading skill is using log scale for any chart longer than ~5 years. On a linear chart, the same percentage move (say, 20%) looks tiny when prices are low and huge when prices are high. This makes long-term linear charts look like rocket ships taking off at the end — a cosmetic illusion that has no analytical content.

On a logarithmic chart, equal percentage moves take up equal vertical space. A doubling from \$10 to \$20 looks the same as a doubling from \$100 to \$200. This is the only way to compare returns across decades or to see whether recent moves are actually larger than historical ones.

Rule: any chart spanning more than ~3–5 years should be on log scale by default. If you only have linear, mentally adjust by asking what the percentage move was, not the dollar move.

4.2 Drawdowns

A drawdown is the percentage decline from a previous peak. It is the most honest measure of “what losing felt like” for that asset. A 50% drawdown requires a 100% gain to recover. A 75% drawdown requires a 300% gain. A 90% drawdown requires a 900% gain.

Reference drawdowns (1996–2026)

Index / Asset

Peak

Trough

Drawdown

Time to recover (peak-to-peak)

S&P 500 (dot-com)

Mar 2000

Oct 2002

~49%

~7 years (2007)

NASDAQ Composite (dot-com)

Mar 2000

Oct 2002

~78%

~15 years (2015)

S&P 500 (GFC)

Oct 2007

Mar 2009

~57%

~5.5 years (2013)

S&P 500 (COVID)

Feb 2020

Mar 2020

~34%

~5 months

S&P 500 (2022)

Jan 2022

Oct 2022

~25%

~24 months (Jan 2024)

NASDAQ-100 (2022)

Nov 2021

Oct 2022

~33%

~18 months

Bitcoin (post-2017)

Dec 2017

Dec 2018

~84%

~3 years

Bitcoin (post-2021)

Nov 2021

Nov 2022

~77%

~2 years

Internalizing this table is more valuable than memorizing any stock pick. When you see a portfolio like this down 30%, you should be able to think: this is roughly the 2022 NASDAQ experience, it took 18 months to recover, the recovery rewarded people who kept buying.

4.3 Rolling returns

Point-to-point returns (e.g., “the S&P 500 returned 10% in 2024”) are noisy. Rolling returns smooth that out by computing the return for every overlapping window of a given length. A 10-year rolling return chart for the S&P 500 shows that since 1926, the index has had positive 10-year nominal returns in roughly 95% of windows, and positive 20-year nominal returns in 100% of windows. That is the empirical foundation for “time in the market beats timing the market.”

4.4 Volatility regimes

Volatility is not constant. Markets cluster: calm periods are followed by calm, storms by storms. The VIX index measures expected 30-day S&P 500 volatility implied by options prices. Long-term average is ~19. Sustained periods below 15 (2017, 2024) signal complacency; sustained periods above 30 (2008, 2020 March) signal panic. Reading the VIX alongside the index itself is core literacy.

4.5 Valuation overlays

Two widely used valuation measures for the S&P 500:

P/E ratio (trailing earnings): Long-term median ~16. >25 is expensive, <12 is cheap, with all the usual caveats about cycles and accounting.

CAPE / Shiller P/E (cyclically adjusted): inflation-adjusted price divided by 10-year average inflation-adjusted earnings. Long-term median ~17. Historically, CAPE ratios above 30 have preceded weaker 10-year forward returns, although timing on this is notoriously poor. Maintained at <https://shillerdata.com>.

Valuation is not a timing tool. It is a probability shifter. High valuation does not mean stocks will fall this year; it means the distribution of 10-year forward returns has historically been narrower and lower.

4.6 Breadth

Breadth measures how many stocks are participating in a move. When the S&P 500 hits a new high but only 30% of its components are above their 200-day moving average, that is a narrow rally and historically more fragile. The 2024–2025 rally has been notably narrow at times, driven heavily by a small number of mega-cap AI beneficiaries.

5. The Past 30 Years — Phase by Phase

This is the narrative spine of the curriculum: 1996 to early 2026, broken into phases. Every phase covers the macro setting, what the major indices did, and what was happening structurally beneath the price action.

5.1 1996–1999 — The Internet Buildout

By 1996 the S&P 500 had been rising for 14 years. Greenspan delivered his “irrational exuberance” speech in December 1996 with the S&P near 740. The S&P would more than double from there before peaking in March 2000. The NASDAQ would 4x.

Macro

Strong U.S. growth, falling inflation, falling unemployment, productivity acceleration from PC and internet adoption.

Asian Financial Crisis (1997) and Russia / LTCM crisis (1998) caused brief sharp drawdowns but did not derail the U.S. bull market.

The Fed cut rates aggressively in late 1998 to backstop the LTCM unwind, which many later argued helped fuel the late-1990s blowoff.

Structural

The internet went from research network to consumer infrastructure. Netscape IPO'd in 1995, Amazon in 1997, Google was founded in 1998.

Telecom build-out: Cisco, Lucent, JDS Uniphase, Nortel — the “picks and shovels” of internet infrastructure — became some of the most valuable companies in the world.

Y2K spending surge front-loaded enterprise IT capex into 1998–1999.

What “normal” looked like at the end of this phase

By March 2000, the NASDAQ Composite peaked above 5,000. CAPE on the S&P 500 hit 44, the highest in U.S. history at that time. Cisco briefly became the most valuable company in the world at a P/E above 200. None of this looked unsustainable to most market participants in real time.

5.2 2000–2002 — The Dot-Com Crash

Between March 2000 and October 2002, the NASDAQ Composite fell ~78%. The S&P 500 fell ~49%. This was the deepest U.S. equity drawdown since the 1973–1974 bear market and the second-worst since 1929. The pain was concentrated in technology and telecom.

How it unfolded

March 2000: NASDAQ peaks. Several flagship dot-coms had already started rolling over.

2000–2001: enterprise IT spend collapses post-Y2K. Telecom carriers built more capacity than the actual internet was using. The fiber glut was real and lasted years.

Sept 2001: 9/11. Markets close for four days, reopen down sharply. The recession was already underway.

2001–2002: a wave of accounting frauds (Enron, WorldCom, Tyco, Adelphia) destroyed remaining confidence in late-cycle tech and telecom names.

October 2002: bottom. NASDAQ at ~1,114, down 78% from the peak.

What survived and what didn't

Survivors that compounded for decades: Microsoft, Oracle, Amazon, Apple, Adobe, Intuit.

Permanently impaired: Cisco (still below its 2000 high in real terms in 2026), Sun Microsystems, Lucent, Nortel, JDS Uniphase, hundreds of dot-coms.

Lessons that survived the cycle

Picking the right industry is not the same as picking the right stocks. The internet was as transformative as the bulls said. Most internet stocks lost 90%+.

Valuation matters at extremes. Buying Cisco at 200x earnings in March 2000 produced negative real returns for 25+ years. Buying Microsoft at 30x earnings produced great long-term returns.

Concentration in a single sector at a peak is the highest-cost mistake in investing history.

5.3 2003–2007 — The Housing-Driven Bull

From the October 2002 bottom to the October 2007 peak, the S&P 500 returned roughly 100% (about 15% annualized). The bull market was led by financials, energy, materials, and emerging markets, not by U.S. tech. The NASDAQ rebounded but was still well below its 2000 peak at the 2007 top.

Macro

Fed cut rates to 1% in 2003 and held them low into 2004, fueling a housing boom.

China's WTO accession (2001) and the commodity supercycle drove energy and materials.

Subprime mortgage origination, securitization, and synthetic CDOs scaled from a niche to a systemic position by 2006–2007.

Tech in the background

Apple launched the iPod (2001), iTunes Store (2003), and the iPhone (2007). Google IPO'd in 2004. Facebook was founded in 2004 and opened to the public in 2006. AWS launched in 2006. The platforms that would dominate the 2010s were being built while the market's attention was on housing.

5.4 2007–2009 — The Global Financial Crisis

The S&P 500 fell ~57% from October 2007 to March 2009, the deepest U.S. equity drawdown since the Great Depression. Unlike 2000, this was not primarily a stock-market crisis; it was a banking and credit crisis that pulled the equity market down with it.

How it unfolded

2007: Bear Stearns hedge funds collapse (June). Subprime origination peaks and rolls over. Bank of England rescues Northern Rock (Sept).

March 2008: Bear Stearns rescued in fire sale to JPMorgan, backstopped by the Fed.

Sept 2008: Lehman fails (Sept 15). AIG bailed out (Sept 16). Money market funds break the buck. Commercial paper market freezes.

Oct 2008–Mar 2009: TARP, ZIRP, QE1, automaker bailouts. S&P bottoms at 666 on March 9, 2009.

Why it matters for an investor with this horizon

If you had bought the S&P 500 at the October 2007 peak and held until early 2026, your annualized return is roughly 10–11% — above the long-run average — despite buying at the worst possible moment in a generation. Time-in-market beat timing for almost every realistic timing rule. This is the most important historical pattern in this entire document.

5.5 2010–2019 — The QE-Era Bull and the FAANG Decade

From March 2009 through December 2019, the S&P 500 returned roughly 17% annualized (with reinvested dividends). The NASDAQ-100 returned roughly 21% annualized. This was the longest U.S. bull market on record.

Macro

Fed funds at 0–0.25% from Dec 2008 to Dec 2015. Three rounds of QE.

Inflation persistently below 2% target despite full employment by mid-decade. The pre-pandemic decade was defined by low inflation and a slow Fed normalization that started in late 2015.

European debt crisis (2010–2012) and Chinese growth scare (2015–2016) caused intra-cycle drawdowns of 15–20% but did not end the bull market.

Structural — the FAANG decade

Smartphones went from <20% to ~80% of global handsets. iOS and Android won.

Cloud computing went mainstream. AWS, Azure, and Google Cloud became core infrastructure.

E-commerce, digital advertising, streaming, and SaaS all hit scale.

By the end of 2019, the five largest U.S. companies (AAPL, MSFT, GOOG, AMZN, FB) were nearly 18% of the S&P 500 — the highest top-5 concentration in decades.

5.6 2020 — The COVID Crash and V-Shaped Recovery

The S&P 500 fell ~34% in 33 trading days (Feb 19 to Mar 23, 2020), the fastest bear market in history. It then recovered the entire loss by August 2020 and ended the year up 16%. The NASDAQ-100 ended 2020 up 47%. This is the single most important episode for understanding modern policy response.

Policy response

The Fed cut to 0–0.25% (March 15) and announced unlimited QE (March 23). It also opened lending facilities for corporate bonds, money markets, and municipalities.

Congress passed the CARES Act (\$2.2T) in March, with subsequent rounds totaling over \$5T in fiscal support.

This combined monetary + fiscal response was an order of magnitude larger and faster than 2008.

What you should take from 2020

Markets are forward-looking. The S&P recovered before the public health crisis was anywhere near resolved, because policy response and earnings forecasts shifted.

The investors who panic-sold in March 2020 missed one of the best 12-month windows in market history.

The 2020–2021 rally set up the 2022 inflation problem. Causality matters.

5.7 2021–2022 — The Inflation Shock and the Rate Hike Cycle

Inflation rose from 1.4% (Jan 2021) to 9.1% (June 2022) — the highest CPI print since 1981. The Fed went from “transitory” in 2021 to the most aggressive hiking cycle since Volcker, raising rates from 0–0.25% to 5.25–5.50% by mid-2023.

How equities responded

S&P 500 peaked Jan 2022, bottomed Oct 2022, down ~25%.

NASDAQ-100 peaked Nov 2021, bottomed Oct 2022, down ~33%.

Long-duration assets — unprofitable tech, SPACs, IPOs from 2020–2021, crypto — fell 60–90%.

Bonds had their worst year in modern history. The Bloomberg US Aggregate fell ~13% in 2022, the worst calendar year on record.

The 2022 lesson

The standard 60/40 stock/bond portfolio (the ballast of a generation of asset allocation) had its worst year of the modern era, because rising rates hurt both halves at the same time. This is a regime that 60/40 is structurally bad at handling and has not been seriously tested in the careers of most working investors.

5.8 2023–2026 — The AI-Led Rally

For a recent-start investor this is the period during which they have actually invested. It is essential to understand it as a period, not as “normal markets.”

What happened

ChatGPT released Nov 2022. By Q1 2023, NVIDIA reported step-function increases in data center demand.

From Oct 2022 to early 2026, the S&P 500 returned roughly 70–80% cumulatively. The NASDAQ-100 returned over 100%. NVDA returned roughly 9–10x.

Fed paused hiking in mid-2023, held rates at 5.25–5.50% through 2024, and began cutting modestly in late 2024.

Concentration in the top 10 S&P 500 names hit ~38%, the highest since the 1960s.

Why this matters for a behavioral baseline

A post-2022 investing experience has been almost entirely a strong bull market with two relatively brief, shallow drawdowns. The resulting sense of what “normal” feels like is calibrated by an unusually good period. Chapter 11 (Behavioral Finance) addresses what happens when you eventually meet a long, grinding drawdown. Be honest with yourself: the 2022 NASDAQ drawdown lasted roughly 12 months. The dot-com drawdown lasted 30 months and didn’t fully recover for 15 years. Both can happen again.

6. Case Studies — The How and Why

Phase narratives give you the shape; case studies give you the texture. These are deep-dives on specific episodes, chosen because they map directly to a portfolio like this.

Case Study A: Cisco — The Right Industry, the Wrong Price

Cisco Systems was the most valuable company in the world for a brief moment in March 2000, with a market capitalization above \$500B and a P/E ratio above 200. By 2002 it had fallen ~85%. Today (2026), Cisco’s nominal price is back near — but still below — its 2000 peak, more than 25 years later. In real (inflation-adjusted) terms it remains well underwater.

The product was real. Cisco built the routers that routed the internet. Revenue grew, earnings grew, the company never went broke. The price was the problem. At 200x trailing earnings, the company had to grow at extraordinary rates for a decade to justify the multiple, and even though it grew impressively, it didn’t grow enough.

Translation to a concentrated tech book: every mega-cap in a book like this (NVDA, AMD, AVGO, AMZN) is a real, durable business. Whether they are good investments from any given starting price depends on the price. This is the lesson the post-2009 generation of investors has the least exposure to.

Case Study B: Amazon — The Survivor

AMZN fell ~95% from its 1999 peak to its 2001 trough (from ~\$113 to ~\$5.50, split-adjusted). It then compounded at roughly 27% annualized for the next 25 years to become a multi-trillion-dollar

company. Holders who bought at the 2000 peak and held through the 95% drawdown did fine — but only if they didn't sell.

Two underappreciated facts. First: even after 25 years of compounding, the original 95% drawdown was real and brutal. Many of the people who held to the 2007 high got out. Second: AWS — the highest-margin, most strategically important Amazon business by 2020s — didn't exist in 2000. The thesis that paid off most for AMZN holders is one that didn't exist when the stock was bought.

This is the core long-horizon investing insight: in a small number of cases, durable businesses with real reinvestment runways can absorb terrible drawdowns and still produce extraordinary multi-decade returns. But you cannot identify which 5% of stocks those will be in advance. You can only identify a basket and hold it through the drawdowns.

Case Study C: NVIDIA — Three Crashes Before the Boom

NVDA is now the headline AI stock and one of the largest companies in the world. Its history before 2023 is critical context.

2000–2002 dot-com: NVDA fell ~90%.

2008 GFC: NVDA fell ~80%.

2018 crypto-mining bust: NVDA fell ~56% in Q4 2018.

2022 AI-cycle reset: NVDA fell ~67% from its Nov 2021 peak.

2023–2025: NVDA returned roughly 9–10x.

Each of those drawdowns was the worst in the company's history at the time. Each was followed by a new cycle that produced a new high. None were forecastable in real time. For an NVDA holder, the historical baseline should be: this stock has had four 50%+ drawdowns in its public history, including a 90% drawdown. That is the price of admission.

Case Study D: The Semiconductor Cycle

Semis are cyclical because the industry has a structural mismatch between demand (which can change in months) and supply (which takes 2–3 years to build a new fab). When demand exceeds supply, prices and margins surge and cap-ex floods in. When the new capacity comes online into a softer demand backdrop, prices and margins collapse.

Reference cycles since 2000

2000–2001 (PC and telecom equipment): SOX index down ~80%.

2008–2009 (GFC): SOX down ~60%.

2011–2012 (PC weakness): SOX down ~30%.

2014–2016 (China slowdown, oil collapse): SOX down ~25%.

2018–2019 (memory glut, China trade war): SOX down ~25%.

2022 (post-COVID demand digestion): SOX down ~35%.

In every cycle, the same companies (Intel, AMD, NVIDIA, Applied Materials) fell hardest at the top, looked broken at the bottom, and led the next move up. An investor holding NVDA + AMD + AVGO at ~75% concentration who has not lived through one of these cycles should plan for it.

Case Study E: 1987 Black Monday in Brief (out of scope, but worth knowing)

Note: this is outside our 30-year window but is the canonical reminder that one-day moves in markets can be larger than most people’s mental models allow. On October 19, 1987, the Dow fell 22.6% in a single day. There was no obvious news catalyst. Portfolio insurance and program trading interacted to amplify selling. The market recovered the loss within two years. The lesson: sometimes large moves happen without comprehensible reasons; that does not mean the market is broken or that you should react.

Case Study F: COVID March 2020 — What Real-Time Panic Looks Like

From Feb 19 to Mar 23, 2020, the S&P 500 fell 34% in 33 trading days. The VIX hit 82 (vs. long-term average ~19). Treasury markets briefly seized. Then the Fed announced unlimited QE on Mar 23 and the market bottomed within days. Anyone who sold during the panic and waited for “the dust to settle” missed the recovery. The S&P 500 was up 70% in the next 12 months.

This is the single best modern test case for the rule: do not sell into panics. The set-up was bad enough that the rule looked stupid in real time. It still worked.

7. Industry and Technology Arcs (1996–2026)

Sector leadership rotates. The names that led one decade rarely lead the next. This chapter walks through the technology and industry arcs that shaped the past 30 years, with emphasis on the names you actually own.

7.1 Semiconductors — from PC era to AI era

The semiconductor industry has gone through three eras in the past 30 years.

PC era (1990s–2000s): dominated by Intel (CPUs) and Microsoft (OS), the “Wintel” duopoly. Memory was a commodity dominated by Samsung, Micron, Hynix.

Mobile era (2007–2018): smartphones drove a new wave of fabless leaders — Qualcomm (modems), ARM (architecture), Apple’s in-house designs (A-series chips), and TSMC as the dominant pure-play foundry. Intel missed the mobile transition.

AI era (2018–present): GPU-based parallel compute (NVIDIA), high-bandwidth memory, custom AI accelerators (Broadcom’s ASICs, Google TPUs, Amazon Trainium). NVIDIA and Broadcom (AVGO) are the most direct beneficiaries; AMD is the credible #2 GPU player.

Holding NVDA + AMD + AVGO is a concentrated bet on the AI era continuing. The historical pattern: each new era has had a small number of giant winners, and the leaders of one era are usually not the leaders of the next.

7.2 Software and SaaS

Two waves: the on-premise era (Microsoft, Oracle, SAP) and the cloud era (Salesforce, ServiceNow, Workday, Atlassian, Snowflake, CrowdStrike).

CrowdStrike (CRWD) is part of the cybersecurity sub-sector — a structural beneficiary of cloud, remote work, and a rising attack surface. Notable inflection: the July 2024 CrowdStrike outage caused one of the largest IT incidents in history and the stock fell ~40% in days. It mostly recovered within months. Worth studying as a real-time example of single-stock event risk in a held position.

7.3 Internet platforms and e-commerce

AMZN is the longest-running e-commerce platform story. Two underappreciated arcs:

Amazon Web Services (launched 2006). For most of the 2010s, AWS contributed the majority of Amazon’s operating income despite being a fraction of revenue.

Advertising. Amazon’s ad business, mostly invisible from the outside, became one of the largest digital ad businesses in the world by the early 2020s, behind only Google and Meta.

This is a core pattern: durable platform companies often turn into something different from what investors originally bought. The thesis updates.

7.4 Financials, energy, and the indices that under-cover them

If you only watched the NASDAQ, you would have missed the entire 2003–2007 bull market and most of the 2022–2023 energy outperformance. The S&P 500 captures these. The Dow captures them. The NASDAQ does not, materially.

8. Bitcoin and Crypto in a Long-Horizon Portfolio

The ~5% crypto sleeve in this profile is FBTC (Fidelity Wise Origin Bitcoin Fund), the spot Bitcoin ETF launched in January 2024. This chapter is the framework for thinking about that allocation.

8.1 What FBTC actually holds

FBTC holds spot Bitcoin in cold storage with Fidelity Digital Assets as custodian. When an investor buys a share, the fund buys Bitcoin. There is no futures roll, no synthetic exposure. The expense ratio is 0.25%. The price tracks Bitcoin closely with small tracking error from fees and rebalancing.

8.2 Bitcoin's historical pattern

Bitcoin trades roughly in 4-year cycles, loosely correlated with the network's halving events (every ~210,000 blocks, roughly every 4 years). Each cycle has produced a new all-time high followed by a 70–90% drawdown. As of early 2026, Bitcoin has had four such cycles.

2011: ~\$1 to ~\$30, then to ~\$2.

2013: ~\$2 to ~\$1,200, then to ~\$200.

2017: ~\$200 to ~\$19,800, then to ~\$3,200.

2021: ~\$3,200 to ~\$69,000, then to ~\$15,500.

2024–2025: new all-time high above \$100,000.

Each of those drawdowns took Bitcoin out of public conversation for 1–3 years. Each was followed by a new cycle. There is no guarantee that the next cycle follows the same pattern.

8.3 Sizing and behavior

The defensible argument for a small Bitcoin allocation in a long-horizon portfolio is asymmetric optionality: in scenarios where it succeeds as a non-sovereign reserve asset, the upside is large; in scenarios where it fails or stagnates, a small allocation is recoverable. The required behavior to make this work is to size it so that a total wipeout would not change your long-run plan, and to actually hold through a 70–80% drawdown without selling.

A 5% allocation is generally inside that envelope for an investor with this horizon and human capital. Two practical rules: (1) re-balance back to target after large moves rather than letting it become 30% of the portfolio at a peak, and (2) never use leverage on a volatile, asymmetric asset.

8.4 Ric Edelman on crypto — a balanced read

Ric Edelman has been notably more bullish on crypto than the mainstream financial-advice industry, advocating allocations of 1–5% (or higher in some interviews) to most retail investors. His arguments are coherent: long horizon, asymmetric upside, low correlation in some regimes. The counter case is that Bitcoin's history is short (~15 years of price data), it has not been tested through a true U.S. recession with high unemployment, and most academic factor work cannot include it because the dataset is too short. A reasonable synthesis: Edelman is not wrong to suggest a small allocation; he may be too confident about the size of the upside relative to the size of the downside; either way, the position-sizing discipline matters more than the directional view.

9. Index Funds vs Actively Managed Funds (and Mutual Funds vs ETFs)

This chapter expands the earlier explanation.

9.1 The structural difference

Index fund: mechanically replicates a published index (S&P 500, NASDAQ-100, total market). No human stock selection. Expense ratios typically 0.02–0.10% per year. Low turnover, low tax drag in taxable accounts.

Actively managed fund: a portfolio manager (and team) selects holdings, hoping to beat the benchmark net of fees. Expense ratios typically 0.50–1.50% per year. Higher turnover, more taxable distributions in non-Roth accounts.

9.2 The data, with sources

The S&P SPIVA U.S. Scorecard is the standard reference. Headline findings (consistent across most years 2010–2024):

Over 1-year periods, roughly 50–60% of active U.S. large-cap funds underperform the S&P 500 (highly variable year to year).

Over 5-year periods, roughly 75–85% underperform.

Over 15-year periods, roughly 85–90% underperform.

Over 20-year periods, the underperformance rate is even higher.

These figures are after fees. They include survivorship bias adjustments — i.e., when a bad fund is closed and merged out of existence (which happens often), SPIVA accounts for it. Without that adjustment, headline stats look better than they should.

9.3 Why active is so hard

Fees compound. 1% extra fees per year over 40 years cuts your final balance by ~30%. This is the largest single factor.

Markets are reasonably efficient. To beat a market, you need to be more right than the marginal informed buyer/seller in that market — hedge funds, prop desks, sell-side analysts, etc. — net of your costs. That is a very high bar.

Sharpe's arithmetic. The total dollar-weighted return of all active investors equals the market return minus their costs. So in aggregate, active management mathematically must underperform passive after fees. Some funds will beat the market in any given year; identifying them in advance is the hard part.

9.4 Mutual funds vs ETFs

This is a separate axis from active vs passive.

Mutual funds: priced once per day at NAV, traded through the fund company. Some have minimums. Generally less tax-efficient than ETFs in taxable accounts because of how internal trading creates taxable distributions to all holders.

ETFs: trade intraday like stocks. Use an in-kind creation/redemption mechanism that almost eliminates taxable distributions in most cases. Generally more tax-efficient in taxable accounts.

In a Roth IRA, the tax efficiency difference is irrelevant (the account is tax-free). In a taxable brokerage, ETFs are usually preferable for buy-and-hold positions. SWPPX is a mutual fund; the equivalent ETFs are VOO, IVV, and SPLG.

10. DCA vs Lump Sum vs Market Timing

This is a question many investors raise. The correct answer depends on which question you are actually asking.

10.1 The three different decisions hiding in this question

Question 1: I have a lump sum today. Should I invest it all at once, or spread it out (DCA)?

Question 2: I am earning an income. Should I keep buying every paycheck regardless of what the market is doing?

Question 3: Should I try to time entries and exits based on what I think the market will do?

These are different questions with different answers.

10.2 Question 1 — Lump sum vs DCA, the data

Vanguard's 2012 study ("Dollar-cost averaging just means taking risk later") and many follow-ups: in roughly two-thirds of historical periods, investing a lump sum immediately produced higher final wealth than DCA'ing the same amount over 6–12 months. The reason is simple — markets go up most of the time, so on average you are better off in the market sooner.

DCA does reduce the worst-case outcome. If you happen to hit a peak with your lump sum, DCA softens that. But the average and median outcome favor lump sum.

Practical synthesis: if you have a lump sum and a long horizon, lump sum has the better expected return. If you know yourself well enough to know that lump-summing into a peak (which sometimes happens) would shake your discipline and cause you to sell, DCA is a reasonable behavioral concession.

10.3 Question 2 — Ongoing DCA from income

If you are investing each paycheck or each month from earned income, you are not really doing DCA in the lump-sum sense. You are doing the only thing you can do — investing money as you earn it. This is the right default. Set up automatic contributions to a Roth IRA and brokerage and let them run.

10.4 Question 3 — Market timing

The data on this is brutal. JPMorgan’s annual “Guide to Retirement” shows that missing the 10 best days in the S&P 500 over a 20-year period roughly cuts your annualized return in half. Missing the 30 best days turns 20-year positive returns into roughly flat. The best days are clustered with the worst days — most of them happen during volatile bear markets. The investors who sell in panics are statistically the most likely to miss the rebounds.

Operational rule: do not sell what you intended to hold long-term because of forecasts. The cost of being wrong about a forecast in either direction is asymmetric.

11. Portfolio Theory and Factor Investing

For the full version: This chapter is the conceptual map; the reading list (Chapter 14) has the books that actually teach it.

11.1 The starting point — Modern Portfolio Theory (Markowitz, 1952)

Markowitz showed mathematically that you can combine assets with the same expected return and reduce total volatility by holding assets that don’t move together. The set of portfolios that maximize expected return for any given volatility is the “efficient frontier.” For a given investor risk tolerance, there is one optimal portfolio on that frontier.

This is the foundation, but it has well-known limits: it assumes returns are normally distributed (they aren’t — fat tails), assumes correlations are stable (they aren’t — they spike during crises), and assumes you can estimate expected returns (you can’t, very well).

11.2 The Capital Asset Pricing Model (Sharpe, 1964)

Adds a risk-free asset and shows that all rational investors should hold the same risky portfolio (the “market portfolio”) and adjust risk by mixing it with the risk-free asset. CAPM gave us beta as the measure of systematic risk and the idea that only systematic risk is rewarded — idiosyncratic risk can be diversified away. CAPM is broken empirically (too simple), but the conceptual scaffolding is foundational.

11.3 The Fama-French factor revolution

Eugene Fama and Kenneth French showed in the 1990s that beta alone does not explain stock returns. Two other factors matter:

Size: small-cap stocks have historically returned more than large-caps over very long periods (the SMB factor).

Value: cheap stocks (low P/B, low P/E) have historically returned more than expensive ones (the HML factor).

This grew into the five-factor model (adding profitability and investment) and the broader factor zoo — momentum (Jegadeesh-Titman), quality, low volatility, and others. Most academic explanation of return differences is now factor-based.

How retail investors access factors

Total market index funds capture the market factor.

Small-cap and value index funds (e.g., VBR, AVUV, DFAV) tilt toward the size and value factors.

Momentum and quality ETFs exist (MTUM, QUAL) but tend to have higher costs and less academic support than size/value.

A reasonable academic-evidence-based core would be: a total U.S. market or S&P 500 index fund (such as SWPPX), plus a small-cap value tilt and an international index fund. Whether to add factor tilts depends on whether you can hold them through long stretches of underperformance — the size and value factors have had multi-year underperformance windows, including 2010–2020.

11.4 Asset allocation — the most important decision

Brinson, Hood, and Beebower (1986) argued that asset allocation explains the majority of the variation in portfolio returns over time. Later researchers debated the exact percentage, but the practical conclusion stands: the mix of stocks/bonds/cash/alternatives matters more than which specific stocks you pick.

For a 40+ year horizon, no near-term liabilities, and stable income, the mainstream recommendation is 80–100% equities. A 95% equities + 5% Bitcoin mix is inside that range. Bonds add ballast in normal regimes but were genuinely terrible in 2022; the case for a young investor holding a meaningful bond allocation has eroded somewhat. Cash earning 4–5% in a money-market fund is a reasonable emergency reserve but is not a long-term allocation.

11.5 Putting it together — a sample academic-leaning portfolio

Not a recommendation, but a useful reference for how the academic literature would suggest building a long-horizon portfolio:

60–70% U.S. total market or S&P 500 (e.g., SWPPX, VTSAX, VOO).

10–20% international developed and emerging market index (e.g., VXUS, IXUS).

5–10% small-cap value tilt (e.g., AVUV).

0–5% Bitcoin or other diversifying asset.

0–10% individual stocks for personal interest, capped so a total wipeout doesn't derail the plan.

This is roughly the inverse of the profile's current allocation. There is no rule that says you have to look like this. There is a strong rule that says you should know how far a portfolio is from this and why.

12. Behavioral Finance — Surviving Your First Drawdown

This chapter is the most important one in the document for an investor early in their journey.

12.1 The behavioral gap

Morningstar's annual "Mind the Gap" study finds that the average investor in U.S. equity funds earns roughly 1–2 percentage points per year less than the funds themselves over long periods. The gap exists because investors buy after rallies and sell after drawdowns. Compounded over 40 years, a 1.5% gap turns into roughly 45% less ending wealth.

12.2 The biases that cause it

Recency bias: weighting recent events too heavily. After a long bull market, investors expect more bull market. After a crash, investors expect more crash. Neither is a forecast.

Loss aversion: psychological pain of a loss is roughly 2x the pleasure of an equal-sized gain (Kahneman-Tversky). Drives investors to sell at lows.

Herdning: it feels safer to be wrong with the crowd than alone.

Overconfidence: investors who have done well in good markets often attribute that to skill rather than to the market environment.

12.3 The pre-commitment device

The single most effective behavioral intervention is to write down, in advance, what you will do in different scenarios. Examples to write before any drawdown:

"If the S&P 500 falls 20%, I will: keep buying my regular contributions and not sell. I will not check my account more than once per week."

"If NVDA falls 50%, I will: re-read my original thesis. If the thesis is intact, I do nothing. If the thesis is broken, I trim to a position size I can hold."

“If Bitcoin falls 70%, I will: not add new money beyond my regular allocation. I will not sell.”

Once you have written this in a calm market, treat it as binding. The future you in a drawdown will want to ignore it. That is exactly why you wrote it.

12.4 What good behavior looks like across the past 30 years

1996–2000 buildup: don’t bet the house at peaks. Some cash for ammo helps.

2000–2002 crash: keep contributing. Don’t double down on the names that lost 90%; rebalance into the survivors.

2003–2007 bull: don’t chase. Keep contributing.

2007–2009 GFC: do not sell at the bottom. Investors who held through and kept buying captured one of the best 10-year windows in history.

2010–2019 bull: don’t panic-sell on every wobble (Aug 2011, Aug 2015, Q4 2018).

2020 COVID: do not sell into the panic. Keep contributing.

2022 rate-hike drawdown: keep contributing. Don’t sell long-duration tech at the trough.

2023–2026 rally: don’t mistake luck for skill. Don’t increase concentration into euphoria.

13. Account Mechanics — Roth IRA vs Brokerage

This profile assumes a Roth IRA and a taxable brokerage. They have very different tax treatment, and the order in which they are filled matters.

13.1 Roth IRA

Funded with after-tax dollars.

Grows tax-free.

Qualified withdrawals after age 59½ (and 5+ years) are tax-free.

2026 contribution limit: \$7,000/year (subject to earned income).

Income limits apply (phase-out starts around \$150K single in 2026; check current numbers).

With a 40+ year horizon, the Roth is mathematically the most valuable account you will ever own. Every dollar of growth in it is permanently tax-free. Maxing it should be the default in any year you have earned income.

13.2 Taxable brokerage

Funded with after-tax dollars.

Long-term capital gains (held >1 year) taxed at 0%, 15%, or 20% based on income.

Short-term capital gains (held ≤1 year) taxed at ordinary income rates.

Dividends taxed annually.

Tax efficiency matters here. ETFs are usually better than actively managed mutual funds because of how they handle internal trading. Holding investments >1 year matters — the difference between short-term and long-term cap gains is a major tax.

13.3 Suggested funding order

1. Emergency fund: 3–6 months of essential expenses in a high-yield savings or money market.
2. Roth IRA: max each year (\$7,000 in 2026).
3. 401(k) up to employer match (if you have one): free money.
4. Remaining 401(k) capacity (Roth or traditional) up to limit.
5. HSA if you have a high-deductible health plan.
6. Taxable brokerage with anything left.

13.4 What to put where (asset location)

In a Roth IRA, prefer your highest-expected-return assets (since the growth is tax-free): individual stocks, growth-tilted index funds, Bitcoin/FBTC. In a taxable brokerage, prefer tax-efficient assets you intend to hold long term: broad index ETFs (VOO, VTI), municipal bonds if you ever hold bonds. Reverse is also defensible — there is academic debate. The strong rule: high-turnover, high-distribution assets should not live in a taxable account if you can help it.

14. Curated Reading List

Organized by category, with the order I'd suggest. The first book in each section is the highest-priority starting point.

14.1 Foundations

“The Little Book of Common Sense Investing” — John C. Bogle. The clearest case for low-cost index investing, from the founder of Vanguard. Read this first.

“A Random Walk Down Wall Street” — Burton Malkiel. The canonical introduction to efficient markets and index investing.

“The Four Pillars of Investing” — William Bernstein. Theory + history + psychology + business in one book. Excellent second read.

“Stocks for the Long Run” — Jeremy Siegel. The empirical case for equities over multi-decade horizons. Skim, don’t read every word.

14.2 Behavioral and psychological

“The Psychology of Money” — Morgan Housel. Short, well-written, the right mindset book.

“Thinking, Fast and Slow” — Daniel Kahneman. The foundation text on behavioral biases.

“Your Money and Your Brain” — Jason Zweig. Behavioral finance applied to investors.

14.3 Portfolio theory and factor investing

“The Intelligent Asset Allocator” — William Bernstein. Practical asset allocation grounded in MPT.

“Expected Returns” — Antti Ilmanen. Dense, comprehensive treatment of return drivers across asset classes. Reference, not cover-to-cover.

“Your Complete Guide to Factor-Based Investing” — Larry Swedroe and Andrew Berkin. The retail-friendly factor investing book.

“Asset Management: A Systematic Approach to Factor Investing” — Andrew Ang. Academic-level treatment if you want depth.

14.4 Market history

“Irrational Exuberance” — Robert Shiller. CAPE ratio, bubbles, and the dot-com era from someone who called it.

“When Genius Failed” — Roger Lowenstein. The LTCM story; required reading for understanding leverage and tail risk.

“The Big Short” — Michael Lewis. The 2008 GFC narrative.

“Too Big to Fail” — Andrew Ross Sorkin. Companion to The Big Short, from the institutional perspective.

“Manias, Panics, and Crashes” — Charles Kindleberger. The longest-running survey of financial crises.

14.5 Buffett, Munger, and the long-term view

“The Essays of Warren Buffett” (Cunningham, ed.). The Buffett shareholder letters organized by topic.

“Poor Charlie’s Almanack” — Charlie Munger. Mental models for thinking, including investing.

“The Outsiders” — William Thorndike. Eight unconventional CEOs and the capital allocation lessons. One of the best business books written.

14.6 On crypto

Lyn Alden — “Broken Money.” The most rigorous mainstream-financial treatment of money, currencies, and Bitcoin.

Ric Edelman — “The Truth About Crypto.” Already in your rotation. Read alongside Alden for balance.

Saifedean Ammous — “The Bitcoin Standard.” Bitcoin maximalist; useful as an opposing argument to mainstream skepticism, but read critically.

14.7 Data and ongoing reading

S&P SPIVA U.S. Scorecard (twice yearly). Active vs passive performance data. Free.

Vanguard research papers (vanguard.com/research). Strong asset allocation and behavioral work.

Fidelity’s Quarterly Sector Update. Sector views and historical context.

FRED (Federal Reserve Economic Data, fred.stlouisfed.org). Macro charts and data, free.

Robert Shiller’s online dataset (shillerdata.com). 150+ years of S&P 500 data.

Aswath Damodaran’s NYU site. Best free valuation and corporate finance resource on the internet.

15. A Suggested 12-Week Study Plan

If you want a concrete schedule, here is one. It assumes ~3–5 hours per week.

Week

Focus

Deliverable / Reading

1

Foundations + indices

Chapters 1–3 of this guide. Bogle, “Little Book.”

2

Chart reading

Chapter 4. Open the interactive artifact and replicate each event on log scale.

3

1996–2002 history

Chapter 5.1–5.2. Shiller, “Irrational Exuberance.”

4

2003–2009 history

Chapter 5.3–5.4. Lewis, “Big Short.”

5

2010–2019 + 2020

Chapter 5.5–5.6. Housel, “Psychology of Money.”

6

2021–2026 + Case A–C

Chapter 5.7–5.8 + Cisco/Amazon/NVDA case studies.

7

Semiconductors deep dive

Case D + Industry chapter 7.1. Read 10-K of NVDA and AVGO.

8

Bitcoin and crypto

Chapter 8 + Lyn Alden “Broken Money.”

9

Index vs active

Chapter 9. Pull the latest SPIVA scorecard.

10

DCA and timing

Chapter 10. Pull the JPMorgan “Missing the best days” slide.

11

Portfolio theory + factors

Chapter 11. Bernstein, “Four Pillars.”

12

Behavioral + write your plan

Chapter 12 + 13. Write your pre-commitment document and your asset allocation target.

16. Closing

Two final thoughts.

First: the single highest-value thing you can do at 18 with a 40+ year horizon is automate. Set up automatic Roth IRA contributions on payday. Set up automatic brokerage contributions on payday. Make the buying happen without your involvement so that bull-market complacency and bear-market panic both have less surface area to attack. Almost everything else in this document is a refinement on that foundation.

Second: the markets you have invested in (2023–2026) have been unusually good. The next 40 years will not be. There will be at least one period of multi-year underperformance, at least one ~30%+ S&P drawdown, probably one ~50%+ S&P drawdown, and almost certainly one or more sector wipeouts that affect names you currently own. None of that has anything to do with whether you will compound wealth successfully over 40 years. The only thing that determines that is whether you keep buying through it.

Good luck, and welcome to the long game.

This is an educational curriculum, generalized for publication from a personalized study guide. It is not investment advice. Figures are drawn from widely cited public sources (S&P SPIVA, Morningstar, Robert Shiller's dataset, Jeremy Siegel's data, FRED, and Vanguard/Fidelity research); verify any specific statistic against its named source.